

Counterfactual Latent Representations: A Neurosymbolic Approach

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MOTIVATION

- Sensitive attributes influence screening and hiring decisions.
- Algorithmic hiring systems offer opportunity to reduce stereotyping and discrimination
- **BUT:** they inherit societal biases from their training data.

Question: Counterfactual Fairness

- „How would a person's CV look like if its sensitive attribute(s) were different?“
- „Would this person be treated differently?“
- Requires causal reasoning to generate this person's data in a counterfactual world
- Non-trivial, particularly with **unstructured high-dimensional data**, such as text

Contribution

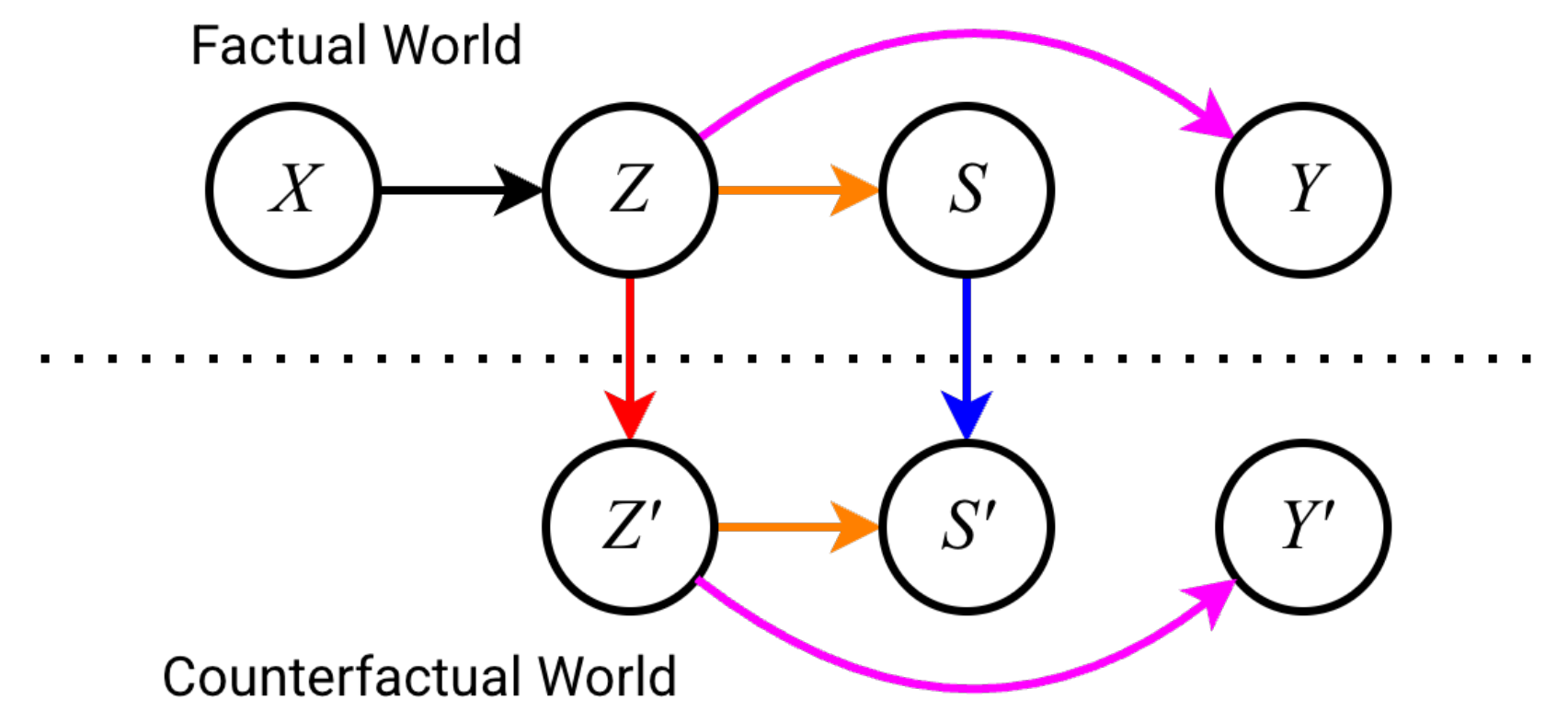
Methodology to generate a counterfactual latent representation of a text input

- **Modular:** Pre-trained VAE to map text to a latent representation + neurosymbolic manipulator model to transform these representations
- **Causal-aware:** Manipulator learns a latent transformation analogously to a causal intervention (abduction-action-prediction) on observable data w.r.t. a sensitive attribute.

PROBLEM FORMULATION

Known :

- X - text input (e.g., a CV)
- A - sensitive attribute (e.g., gender)
- Z - latent representation produced by a pre-trained encoder $Z = f(X)$



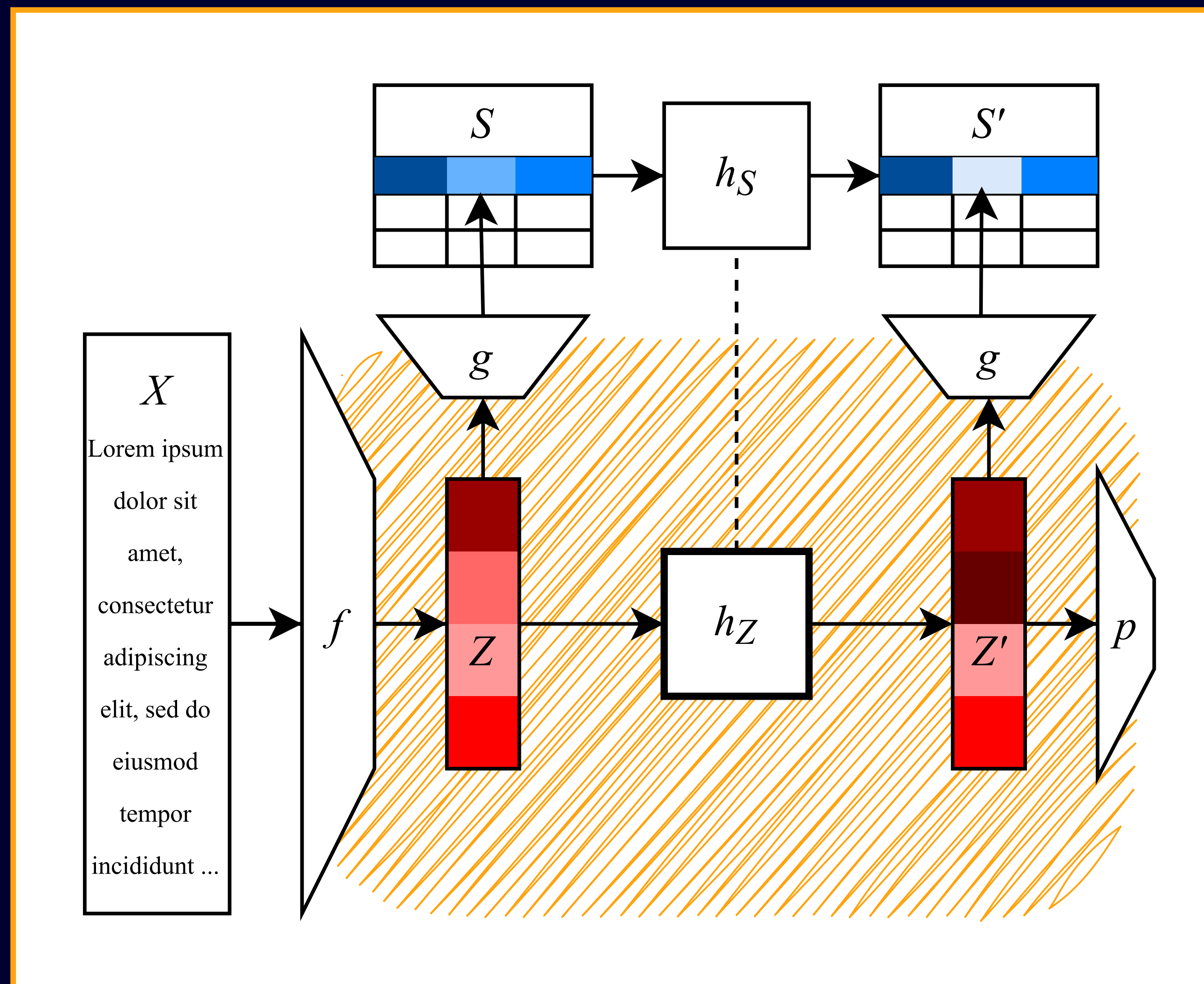
Goal :

- Z' - counterfactual latent representation of X w.r.t. A
- $\rightarrow$ e.g., a CV-representation of the same person if their gender were different

Auxiliary information:

- S - set of bias-relevant observable tabular features (descendants of A in the causal graph, e.g., number of programming languages, years of experience, etc.)
- M - structural causal model over S that captures the causal effect of A on a task.

PROCEDURE



0. Pre-trained encoder $f: X \rightarrow Z$

1. Semantic Decoder $g: Z \rightarrow S$

- Partial mapping from latent space to structured representation S
- Trained on synthetic data with matching X and S
- Consistency signal for step 2

2. Neurosymbolic Manipulator $h_{S/Z}$

- Symbolic operation: $h_S: S \rightarrow S'$ calculates the counterfactual S' to S w.r.t. A via abduction-action-prediction on M
- Neural manipulator: $h_Z: Z \rightarrow Z'$ learns the transformed latent representation $Z' := h_Z(Z)$ analogously to $h_S(S)$

$$h_S(g(Z)) = g(h_Z(Z))$$

- Training objective (d_S is a distance function in S -space):

$$\mathcal{L} = \underbrace{d_S(h_S(g(Z)), g(h_Z(Z)))}_{\text{Consistency Constraint}} + \underbrace{\alpha \cdot \|h_Z(Z) - Z\|_1}_{L_1: \text{Sparsity Penalty}} + \underbrace{\beta \cdot \|h_Z(Z) - Z\|_2^2}_{L_2: \text{Deviation Penalty}}$$

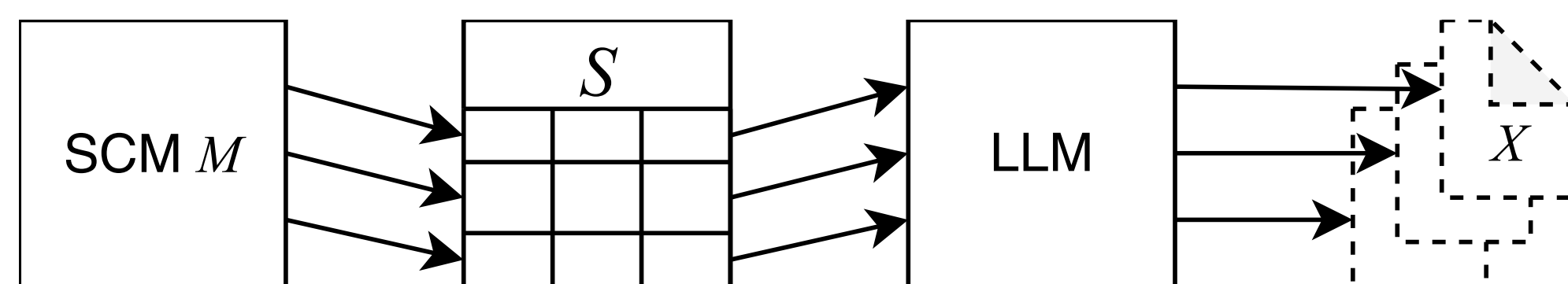
3. Downstream Task

- E.g., a predictor p with constraint $p(Z) \neq p(Z')$

DATA & HYPOTHESES

Synthetic Data

- Approach similar to Toker et al. (2026)
- Dataset with matching X and S to train the semantic decoder g

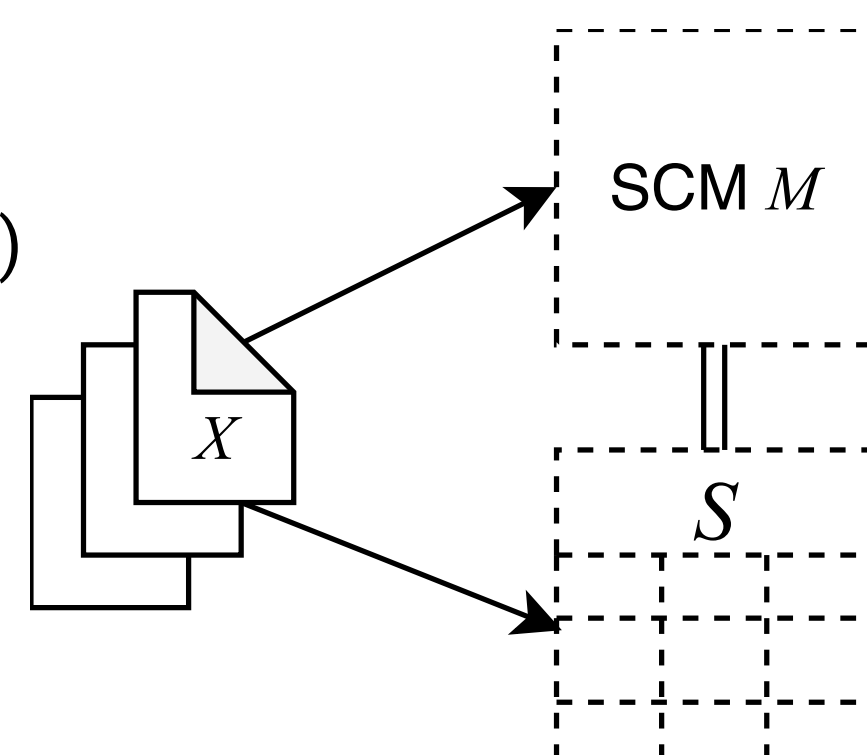


Challenges:

- LLM may introduce additional hidden confounders descending from A
- External validity

Real World Data

- CV dataset, e.g., *Bias in Bios* (De-Arteaga et al. 2019)
- **Challenges:**
 - SCM requires strong causal assumptions
 - Tabular representation S is unknown



Hypotheses

1. **Analogy:** h_Z can be trained to generate the counterfactual of Z analogously to h_S
2. **Sparsity:** h_Z only adjusts observed confounders
3. **Effectiveness:** Z' and Z can be used to debias a prediction model
4. **Meaningfulness:** There exists a decoder that can retrieve the counterfactual X' to X from Z'

RELATED WORK

- **(Path-specific) counterfactual fairness** (Chiappa 2019; Kusner et al. 2017): fairness through interventions on specified causal paths in an SCM.
- **Fair representation learning** (Zemel et al. 2013; Creager et al. 2019; Oh et al. 2022): learning a latent representation that is invariant to sensitive attributes.
- **Deep Structural Causal Models** (Pawlowski et al. 2020): SCMs with VAEs as functional connectors, trained end-to-end on observational data.
- **Neurosymbolic latent reasoning** (Asai et al. 2022): dividing a complex task into a neural „perception“ encoder and a latent symbolic reasoning module.

DISCUSSION

Advantage: Modularity

- Flexible, auditable and composable
- Independence from encoder training

Limitations: SCM Assumptions

- No hidden confounders $\rightarrow M$ is accurate
- Full observability of $S \rightarrow g$ is well-fit

Scope

- Value beyond fairness, e.g., privacy or safety $\rightarrow$ **Latent Symbolic Interventions**

Central open question:

Can a latent manipulator learn a faithful analogue of the symbolic operation in practice?

